1 Periodic functions and Fourier series

1.1 Complex valued functions

Throughout this course we will make extensive use of functions defined over the complex plane, particularly the *complex exponential function* e^z. Before diving into the heavy stuff, let us review the basics of complex functions and remind ourselves how the exponential function is extended from $\mathbb{R}$ to $\mathbb{C}$.

The field of complex numbers $\mathbb{C}$ consists of all numbers $z = x + \mathrm{i}y$, where x and y are real numbers and $\mathrm{i} = \sqrt{-1}$. Visualising $\mathbb{C}$ is easy; we simply draw the plane $\mathbb{R}^2$ and represent a complex number $x+\mathrm{i}y$ by the point $(x, y) \in \mathbb{R}^2$. We refer to this as the *complex plane* or *Argand plane*. Through this identification, the complex plane inherits a norm $|\cdot| : \mathbb{C} \to \mathbb{R}_{\geq 0}$ given by

$$|z| = |x + \mathrm{i}y| = \sqrt{x^2 + y^2}. \tag{1.1}$$

As with any good norm, it satisfies the *triangle inequality*

$$|z_1 + z_2| \leq |z_1| + |z_2|, \qquad \forall z_1, z_2 \in \mathbb{C}. \tag{1.2}$$

We often call $|z|$ the *modulus* of z. Associated to every complex number $z = x + \mathrm{i}y$ is its *complex conjugate* $\bar{z} = x - \mathrm{i}y$. Using this, we have the following formulae for the real and imaginary parts of z

$$\mathrm{Re}(z) = \frac{z + \bar{z}}{2}, \qquad \mathrm{Im}(z) = \frac{z - \bar{z}}{2\mathrm{i}}. \tag{1.3}$$

It is also easy to show that $|z|^2 = z\bar{z}$.

Now suppose we have a function $f : \mathbb{R} \to \mathbb{C}$. We write $f(t)$ as $u(t) + \mathrm{i}v(t)$ where $u(t)$ is the real part of $f(t)$ and $v(t)$ is the imaginary part. Doing so allows us to define the derivative of f as

$$f'(t) = u'(t) + \mathrm{i}v'(t). \tag{1.4}$$

Similarly, if $U(t)$ and $V(t)$ are anti-derivatives of $u(t)$ and $v(t)$ respectively, the integral of f over $[a, b]$ is given by

$$\int_a^b f(t)\,\mathrm{d}t = F(b) - F(a), \quad F(t) \doteq U(t) + \mathrm{i}V(t). \tag{1.5}$$

In other words, by splitting a complex valued function of a real variable into real and imaginary parts, we can do calculus as if we were dealing with two real valued functions.

Example 1.1. Many real functions can be extended to complex valued functions. For example, the exponential function e^x can be extended by noting that its Maclaurin series

$$\mathrm{e}^z = \sum_{n=0}^{\infty} \frac{z^n}{n!}, \tag{1.6}$$

has an infinite radius of convergence and, therefore, is well-defined at any value of $z \in \mathbb{C}$.

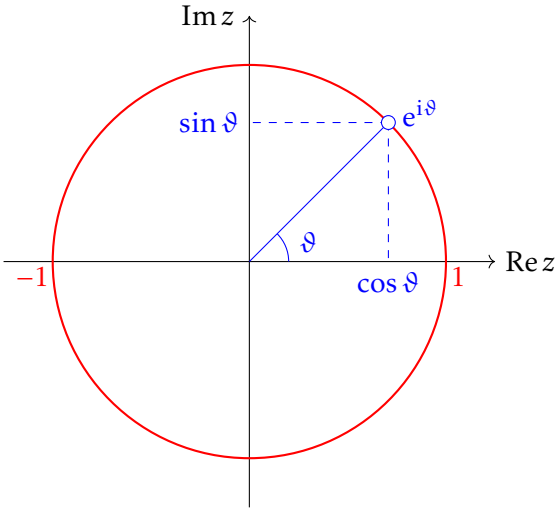


Figure 1: A plot of $z = e^{i\vartheta}$ in the complex plane.

Furthermore, if we set $z = i\vartheta$ with $\vartheta \in \mathbb{R}$, we can split $e^{i\vartheta}$ into real and imaginary parts as

$$\begin{aligned} e^{i\vartheta} = \sum_{n=0}^{\infty} \frac{(i\vartheta)^n}{n!} &= \left(1 - \frac{\vartheta^2}{2} + \frac{\vartheta^4}{24} - \dots\right) + i\left(\vartheta - \frac{\vartheta^3}{6} + \frac{\vartheta^5}{120} - \dots\right) \\ &= \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!}\vartheta^{2k} + i\sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!}\vartheta^{2k+1}. \end{aligned} \tag{1.7}$$

We now recognise the first sum as the Maclaurin series for $\cos\vartheta$ and the second as the Maclaurin series for $\sin\vartheta$. This gives the famous *Euler's formula*

$$e^{i\vartheta} = \cos\vartheta + i\sin\vartheta. \tag{1.8}$$

From this we can immediately deduce that $|e^{i\vartheta}| = 1$, so Euler's formula tells us that if we draw the function $f(\vartheta) = e^{i\vartheta}$ in the complex plane for $\vartheta \in \mathbb{R}$, we will get a circle of radius 1 (as shown in fig. 1).

We can derive several useful identities using Euler's formula. For example, using the fact that $\cos(-\vartheta) = \cos(\vartheta)$ and $\sin(-\vartheta) = -\sin(\vartheta)$, we see that

$$\overline{e^{i\vartheta}} = \cos\vartheta - i\sin\vartheta = \cos(-\vartheta) + i\sin(-\vartheta) = e^{-i\vartheta}. \tag{1.9}$$

Using the formulae for $\mathrm{Re}(z)$ and $\mathrm{Im}(z)$ together with Euler's formula, we obtain the following useful expressions for $\cos\vartheta$ and $\sin\vartheta$

$$\cos\vartheta = \frac{e^{i\vartheta} + e^{-i\vartheta}}{2}, \qquad \sin\vartheta = \frac{e^{i\vartheta} - e^{-i\vartheta}}{2i}. \tag{1.10}$$

As a sanity check, we can substitute these expressions into $\cos\vartheta + i\sin\vartheta$ and recover $e^{i\vartheta}$, as expected.

Example 1.2. (i) As Euler's formula gives us the real and imaginary parts of $e^{i\alpha\vartheta}$, we can calculate the integrals and derivatives as described above. Starting first with the derivative, we have

$$\frac{d}{d\vartheta}\left(e^{i\alpha\vartheta}\right) = -\alpha\sin\alpha\vartheta + i\alpha\cos\alpha\vartheta = i\alpha(\cos\alpha\vartheta + i\sin\alpha\vartheta) = i\alpha e^{i\alpha\vartheta}. \tag{1.11}$$

This also implies that $\frac{1}{i\alpha}e^{i\alpha\vartheta}$ is an anti-derivative of $e^{i\alpha\vartheta}$.

(ii) Now consider the function $e^{in\vartheta}$ with $n \in \mathbb{Z}$. First taking $n \neq 0$, we have

$$\int_0^{2\pi} e^{in\vartheta}\,d\vartheta = \left[\frac{e^{in\vartheta}}{in}\right]_0^{2\pi} = \frac{e^{2\pi in} - e^0}{in} = 0. \tag{1.12}$$

However, if $n = 0$, then $e^{in\vartheta} = 1$ and the integral is simply 2π. In summary

$$\int_0^{2\pi} e^{in\vartheta}\,d\vartheta = \begin{cases} 2\pi & n = 0, \\ 0 & n \neq 0. \end{cases} \tag{1.13}$$

1.2 Periodic functions and complex Fourier series

In many real life problems, we are interested in signals or functions that repeat after some time has elapsed. Electrical signals, audio waves and ocean waves provide just a few examples. Functions that display this property are called *periodic* and will be the focus of the rest of this chapter. Before getting to the interesting stuff, we have to formalise this concept.

Definition 1.1. A function $f : \mathbb{R} \to \mathbb{C}$ is *periodic with period* $T \neq 0$ if

$$f(t + T) = f(t), \tag{1.14}$$

for all $t \in \mathbb{R}$.

This definition implies that if f has period T, it also has period nT for any non-zero integer n, but we will often refer to the minimal period as 'the period of f'.

Example 1.3. We have the following examples of periodic functions:

- The functions $\sin t$ and $\cos t$ are 2π-periodic. By Euler's formula, e^{it} is also 2π-periodic.
- To obtain examples of a T-periodic function for arbitrary T, we can scale $t \mapsto 2\pi t/T$ in the above to get $\sin(2\pi t/T)$, $\cos(2\pi t/T)$ and $e^{2\pi it/T}$. More generally, if $f(t)$ is a T-periodic function, the scaled function $g(t) = f(tT/K)$ is K-periodic.
- There is no requirement that a periodic function needs to be continuous. For example,

we can consider the function $\chi_{\mathbb{Z}}(t)$ defined by

$$\chi_{\mathbb{Z}}(t) = \begin{cases} 1 & t \in \mathbb{Z}, \\ 0 & \text{otherwise.} \end{cases} \tag{1.15}$$

Concretely, it is equal to 1 when t is an integer and is 0 everywhere else. This function is discontinuous at every integer and has period 1. In fact, if you replace $\mathbb{Z}$ by $\mathbb{Q}$ in the above definition, you get a function that is everywhere discontinuous and has infinitely many periods of the form $1/p$ for every prime p.

As we'll see later, periodic functions have some nice properties, but a useful one for now is that integrating over a full period gives the same result regardless of the start point. That is, for any $a \in \mathbb{R}$

$$\int_a^{a+T} f(t)\,\mathrm{d}t = \int_0^T f(t)\,\mathrm{d}t, \tag{1.16}$$

if f has period T. To show this, you can split up the integral into two parts and change variables to shift the integration range to be of the form $[0, T]$.

Motivated by his work on the heat equation, Joseph Fourier asked if arbitrary periodic functions could be written as a series of simpler periodic functions. His claim was bold and somewhat imprecise, but the idea would prove to be revolutionary, leading to the development of *harmonic analysis*. This idea leads to the key definition of this chapter.

Definition 1.2. Let $f : \mathbb{R} \to \mathbb{C}$ be a T-periodic function. The (complex) *Fourier series* of f is given by

$$f(t) \sim \sum_{n\in\mathbb{Z}} c_n \mathrm{e}^{\mathrm{i}n\Omega t}, \qquad c_m = \frac{1}{T}\int_0^T f(t)\mathrm{e}^{-\mathrm{i}m\Omega t}\,\mathrm{d}t, \tag{1.17}$$

where $\Omega = 2\pi/T$ is the *angular frequency*. The numbers c_n are called the *Fourier coefficients*, while the finite series

$$S_N(t) = \sum_{n=-N}^{N} c_n \mathrm{e}^{\mathrm{i}n\Omega t}, \tag{1.18}$$

is called the Nth *partial sum* of the Fourier series.

To guarantee that these integrals exist, we demand that f must be *absolutely integrable* over a single period, meaning that

$$\int_0^T |f(t)|\,\mathrm{d}t < \infty. \tag{1.19}$$

This ensures that each c_n is well-defined since

$$\left|\int_0^T f(t)\mathrm{e}^{-\mathrm{i}n\Omega t}\,\mathrm{d}t\right| \le \int_0^T |f(t)\mathrm{e}^{-\mathrm{i}n\Omega t}|\,\mathrm{d}t = \int_0^T |f(t)|\,\mathrm{d}t, \tag{1.20}$$

therefore bounding the integral for any $n \in \mathbb{Z}$. It is easy to show that absolutely integrable functions form a vector space which we denote by $L^1([0,T])$.

As presented, this is just a definition of some series and we cannot really say how f relates to its Fourier series, so we use $\sim$ as shorthand for 'has Fourier series'. To motivate why this is at least a reasonable definition, let us suppose that $f(t)$ can be written as

$$f(t) = \sum_{n\in\mathbb{Z}} c_n e^{in\Omega t}. \tag{1.21}$$

Then we can multiply by $e^{-im\Omega t}$ and integrate to get

$$\int_0^T f(t)e^{-im\Omega t}\,dt = \sum_{n\in\mathbb{Z}}\left(\int_0^T c_n e^{i(n-m)\Omega t}\,dt\right), \tag{1.22}$$

assuming that we can exchange summation and integration. Now we change variables to $s = 2\pi t/T = \Omega t$ to change our integration range to $[0, 2\pi]$

$$\sum_{n\in\mathbb{Z}}\left(\int_0^T c_n e^{i(n-m)\Omega t}\,dt\right) = \frac{T}{2\pi}\sum_{n\in\mathbb{Z}}\left(\int_0^{2\pi} c_n e^{i(n-m)s}\,ds\right). \tag{1.23}$$

As seen in eq. (1.13), the only integral on the right-hand side that is non-zero is when $n-m=0$ where the integral is equal to 2π. Putting this together we have

$$\int_0^T f(t)e^{-im\Omega t}\,dt = Tc_m, \tag{1.24}$$

recovering the definition of the Fourier coefficients in the definition! So we see that the definition is at least consistent if $f(t)$ is equal to its Fourier series. We will return to the issue of when this is actually true later in this chapter, but for now we will turn a blind eye to it.

Example 1.4. The *sawtooth wave* is a 2π-periodic function defined on $[-\pi,\pi)$ by $f(t) = t$. For $n \neq 0$, the Fourier coefficients are given by

$$\begin{aligned} c_n &= \frac{1}{2\pi}\int_{-\pi}^{\pi} te^{-int}\,dt = \left[-\frac{te^{-int}}{2\pi in}\right]_{-\pi}^{\pi} + \frac{1}{2\pi in}\int_{-\pi}^{\pi} e^{-int}\,dt, \\ &= -\frac{1}{2\pi in}(\pi e^{-in\pi} + \pi e^{in\pi}), \\ &= \frac{(-1)^{n+1}}{in}, \end{aligned} \tag{1.25}$$

where we've used $e^{in\pi} = (-1)^n$ in the final line. Since this has a pole at $n=0$, we must check the $n=0$ case separately

$$c_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} t\,dt = 0. \tag{1.26}$$

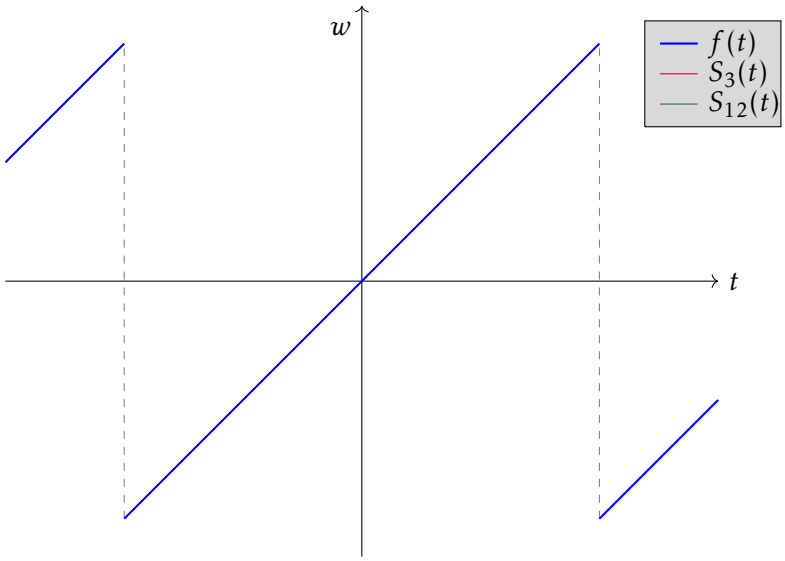


Figure 2: The $N = 3$ and $N = 12$ partial sums of the sawtooth wave compared with the function itself.

Therefore, $f(t)$ has Fourier series

$$f(t) \sim \sum_{n \neq 0} \frac{(-1)^{n+1}}{\mathrm{i}n} \mathrm{e}^{\mathrm{i}nt}. \tag{1.27}$$

Notice that the complex conjugate of this series is itself, showing that it is actually real. Plotting the partial sums for $N = 3$ and $N = 12$ gives fig. 2.

1.3 Trigonometric Fourier series

If we are interested in finding the Fourier series for a real function $f : \mathbb{R} \to \mathbb{R}$, it is often more convenient to try and write it in terms of $\cos(n\Omega t)$ and $\sin(n\Omega t)$. This can be done by using Euler's formula to rewrite the complex form as

$$f(t) \sim \sum_{n=-\infty}^{\infty} c_n \mathrm{e}^{\mathrm{i}\Omega nt} = \sum_{n=-\infty}^{\infty} c_n [\cos(n\Omega t) + \mathrm{i}\sin(n\Omega t)]. \tag{1.28}$$

Since $\cos(-t) = \cos(t)$ and $\sin(-t) = -\sin(t)$, we can pair up the terms in the sum with $n = k$ and $n = -k$ to get

$$\begin{aligned} f(t) &\sim c_0 + \sum_{n=1}^{\infty} [(c_n + c_{-n})\cos(n\Omega t) + \mathrm{i}(c_n - c_{-n})\sin(n\Omega t)], \\ &\doteq \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(n\Omega t) + b_n \sin(n\Omega t), \end{aligned} \tag{1.29}$$

where we've defined $a_n = c_n + c_{-n}$ and $b_n = \mathrm{i}(c_n - c_{-n})$. When f is a real function, it can be easily seen that $c_{-n} = \bar{c}_n$, so we can write $a_n = 2\mathrm{Re}(c_n)$ and $b_n = -2\mathrm{Im}(c_n)$. By substituting in the integral expression for the complex Fourier coefficients, we see that a_n and b_n can be computed via

$$a_n = \frac{2}{T}\int_0^T f(t)\cos(n\Omega t)\,\mathrm{d}t, \qquad b_n = \frac{2}{T}\int_0^T f(t)\sin(n\Omega t)\,\mathrm{d}t. \tag{1.30}$$

Note that the integral formula for a_n is consistent with the factor of $\frac{1}{2}$ multiplying a_0 in the expansion above.

Example 1.5. Returning to example 1.4, we saw that the function defined on $[-\pi, \pi)$ as $f(t) = t$ has complex Fourier coefficients

$$c_n = \begin{cases} \frac{(-1)^{n+1}}{\mathrm{i}n} & n \neq 0, \\ 0 & n = 0. \end{cases} \tag{1.31}$$

From this we see that $c_{-n} = -c_n$ for all n, so we have

$$a_n = c_n + c_{-n} = 0, \qquad b_n = \mathrm{i}(c_n - c_{-n}) = \frac{2(-1)^{n+1}}{n}. \tag{1.32}$$

The Fourier series of $f(t)$ in trigonometric form is therefore

$$f(t) \sim \sum_{n=1}^{\infty} \frac{2(-1)^{n+1}}{n} \sin nt. \tag{1.33}$$

Usually, it is easier to calculate the complex Fourier coefficients first and then use the linear relations to obtain the trigonometric coefficients. However, if f has certain symmetry properties a nice simplification occurs.

Definition 1.3. A function $f : \mathbb{R} \to \mathbb{C}$ is called *even* if

$$f(-x) = f(x), \quad \forall x \in \mathbb{R}. \tag{1.34}$$

Similarly, f is said to be *odd* if

$$f(-x) = -f(x), \quad \forall x \in \mathbb{R}. \tag{1.35}$$

There are several easy examples of odd and even functions. Functions of the form $f(t) = t^n$ are odd when n is odd and even when n is even, while $\cos t$ and $\sin t$ are even and odd respectively. However, note that $\mathrm{e}^{\mathrm{i}nt}$ is neither odd nor even for $n \neq 0$.

Clearly, the sum of two even (resp. odd) functions is also even (resp. odd), but under multiplication we must be more careful. While the product of two even functions is even, if f and g are odd functions, we have

$$f(-x)g(-x) = (-f(x))(-g(x)) = f(x)g(x), \tag{1.36}$$

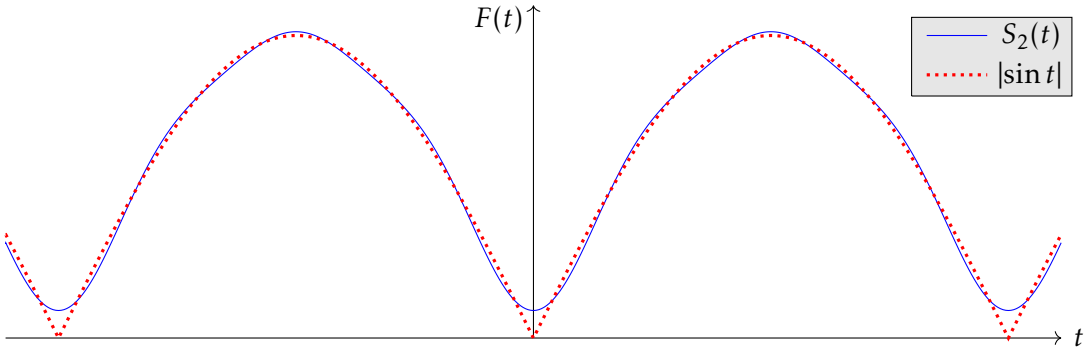


Figure 3: Fourier series with 3 terms for $f(t) = |\sin t|$.

meaning the product is actually even! It is also easy to show that the product of an odd function and an even function is always odd.

Due to the symmetry of odd and even functions under $t \mapsto -t$, integrals over the interval $[-a, a]$ simplify tremendously. Concretely, for any function f we have

$$\int_{-a}^{a} f(t)\,\mathrm{d}t = \int_{-a}^{0} f(t)\,\mathrm{d}t + \int_{0}^{a} f(t)\,\mathrm{d}t = \int_{0}^{a} (f(t) + f(-t))\,\mathrm{d}t, \tag{1.37}$$

where we've used the substitution $t \mapsto -t$ in the integral over $[-a, 0]$. Supposing now that f is either even or odd gives

$$\int_{-a}^{a} f(t)\,\mathrm{d}t = \begin{cases} 2\int_0^a f(t)\,\mathrm{d}t & f \text{ even}, \\ 0 & f \text{ odd}. \end{cases} \tag{1.38}$$

Using this formula, we see that if $f(t)$ is an even T-periodic function, then $f(t)\sin(n\Omega t)$ is odd and the sine Fourier coefficients b_n vanish. Similarly, if $f(t)$ is odd, then so is $f(t)\cos(n\Omega t)$, meaning a_n vanishes! In summary, we have

$$\begin{aligned} f \text{ even:} \quad & b_n = 0 \Longrightarrow f(t) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(n\Omega t), \quad a_n = \frac{4}{T}\int_0^{T/2} f(t)\cos(n\Omega t)\,\mathrm{d}t, \\ f \text{ odd:} \quad & a_n = 0 \Longrightarrow \quad f(t) \sim \sum_{n=1}^{\infty} b_n \sin(n\Omega t), \quad b_n = \frac{4}{T}\int_0^{T/2} f(t)\sin(n\Omega t)\,\mathrm{d}t. \end{aligned} \tag{1.39}$$

This is precisely why the Fourier series for the sawtooth wave only contains sine terms: it is an odd function, so $a_n = 0$ for every n. Furthermore, we can calculate b_n directly as

$$b_n = \frac{2}{\pi}\int_0^{\pi} t\sin nt\,\mathrm{d}t = \left[-\frac{2t\cos nt}{n\pi}\right]_0^{\pi} + \frac{2}{n\pi}\int_0^{\pi} \cos nt\,\mathrm{d}t = \frac{2(-1)^{n+1}}{n}, \tag{1.40}$$

matching what we got before.

Example 1.6. For a more involved example, consider the function $f(t) = |\sin t|$. This is an even function and π-periodic, so its Fourier series contains only terms of the form $\cos(2nt)$. The cosine Fourier coefficients are given by

$$a_n = \frac{4}{\pi}\int_0^{\pi/2} \sin(t)\cos(2nt)\,\mathrm{d}t. \tag{1.41}$$

We can use the product-to-sum identities to get

$$a_n = \frac{2}{\pi}\int_0^{\pi/2} [\sin(1+2n)t + \sin(1-2n)t]\,\mathrm{d}t = \frac{4}{\pi(1-4n^2)}. \tag{1.42}$$

Note that since $n = 0$ is not a pole of a_n, we do not need to evaluate a_0 separately. This gives us the Fourier expansion

$$f(t) \sim \frac{2}{\pi} - \frac{4}{\pi}\sum_{n=1}^{\infty} \frac{\cos(2nt)}{4n^2-1}. \tag{1.43}$$

A comparison between $f(t)$ and the first three terms of the Fourier series is shown in fig. 3.

1.4 Half-range series

Quite often it is useful to obtain a Fourier series for a function $f(t)$ that is only defined on an interval $[0, T/2)$. This often occurs when solving PDEs, for example. In order to do this, we need to extend $f(t)$ to a periodic function over $\mathbb{R}$, but there are several ways to do this. Since we have the freedom to extend it however we like, we can choose the extension to be either odd or even and the Fourier series computation becomes fairly straightforward. Given such an f, we define the *even extension* $F_{\text{even}}(t)$ to be

$$F_{\text{even}}(t) = \begin{cases} f(t) & \text{if } 0 \le t < T/2, \\ f(-t) & \text{if } -T/2 < t < 0. \end{cases} \tag{1.44}$$

Treating this as an even T-periodic function, we can Fourier expand it as in the previous section

$$F_{\text{even}}(t) \sim \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(n\Omega t), \quad a_n = \frac{4}{T}\int_0^{T/2} f(t)\cos(n\Omega t)\,\mathrm{d}t, \tag{1.45}$$

where $\Omega = 2\pi/T$. We call this the *Fourier cosine series* or *half-range cosine series*. Similarly, the *odd extension* of f is defined as

$$F_{\text{odd}}(t) = \begin{cases} f(t) & \text{if } 0 \le t < T/2, \\ -f(-t) & \text{if } -T/2 < t < 0, \end{cases} \tag{1.46}$$

which leads to the *Fourier sine series* or *half-range sine series*

$$F_{\text{odd}}(t) \sim \sum_{n=1}^{\infty} b_n \sin(n\Omega t), \quad b_n = \frac{4}{T}\int_0^{T/2} f(t)\sin(n\Omega t)\,\mathrm{d}t. \tag{1.47}$$

On the interval $[0, T/2)$, where the original function was defined, these two series should agree provided they converge. So we are free to choose which one to work with if we require a half-range series.

Example 1.7. Consider the function $f(t) = t$ on the half-range $[0, \pi)$. We can extend this to an even function and calculate its cosine Fourier coefficients as

$$\begin{aligned} a_n &= \frac{2}{\pi}\int_0^\pi t\cos nt\,\mathrm{d}t = \frac{2}{\pi}\left[\frac{t\sin nt}{n}\right]_0^\pi - \frac{2}{\pi n}\int_0^\pi \sin nt\,\mathrm{d}t, \\ &= \frac{2}{\pi n^2}[\cos nt]_0^\pi, \\ &= \frac{2((-1)^n - 1)}{\pi n^2}, \end{aligned} \tag{1.48}$$

so a_n is non-zero only when n is odd. It is easy to see that $a_0 = \pi$, so the cosine Fourier series is therefore

$$F_{\text{even}}(t) \sim \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2((-1)^n - 1)}{\pi n^2}\cos nt = \frac{\pi}{2} - \sum_{m=0}^{\infty} \frac{4}{\pi(2m+1)^2}\cos(2m+1)t, \tag{1.49}$$

where in the final equality we've introduced $n = 2m + 1$.

1.5 Decay rate of Fourier coefficients

So far we've been calculating Fourier series without really knowing if the series actually captures the original function in question, using $\sim$ to just mean 'has Fourier series'. We now turn to this question, first justifying when these series have a finite value.

Recall from single variable analysis that infinite series can display different behaviours. For example, we have the following.

- The harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent as the partial sums do not tend to a finite limit;
- The alternating harmonic series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ is *conditionally convergent*, meaning that it has a finite limit, but the sum of the absolute values does not converge;
- Finally, the sum $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$ is *absolutely convergent*, meaning both it and the sum of absolute values converge to a finite value.

Our goal is to establish how the Fourier coefficients decay as $n \to \infty$ to see when the Fourier partial sums converge.

Proposition 1.1. Let f be an absolutely integrable T-periodic function with Fourier coefficients c_n. Then

$$\text{(i) } |c_n| \leq \frac{1}{T}\int_0^T |f(t)|\,\mathrm{d}t, \qquad \text{(ii) } \lim_{n\to\pm\infty} c_n = 0.$$

Proof. (i) The triangle inequality for integrals implies that

$$|c_n| \leq \left| \frac{1}{T} \int_0^T f(t) \mathrm{e}^{-\mathrm{i}n\Omega t} \, \mathrm{d}t \right| \leq \frac{1}{T} \int_0^T |f(t) \mathrm{e}^{-\mathrm{i}n\Omega t}| \, \mathrm{d}t = \frac{1}{T} \int_0^T |f(t)| \, \mathrm{d}t, \tag{1.50}$$

which is finite since f is absolutely integrable.

(ii) This is known as the Riemann-Lebesgue lemma and proving it is beyond the scope of the course. □

This proposition tells us that the Fourier coefficients are bounded above and tend to zero as $n \to \infty$, but this isn't enough for convergence. For example, every term of the divergent harmonic series is bounded by 1 and approaches zero as n grows, so we need a stronger result to ensure convergence. The natural next step is to impose more structure on f.

Proposition 1.2. Let f be a continuous T-periodic function that is k-times continuously differentiable with $k \geq 1$. Then there exists an $M > 0$ such that

$$|c_n| \leq \frac{M}{|n|^k}, \quad \forall n \neq 0, \tag{1.51}$$

where c_n are the Fourier coefficients of f.

Proof. By hypothesis $f^{(k)}$ is T-periodic and continuous, so $f^{(k)} \in L^1([0,T])$. We can therefore Fourier expand $f^{(k)}$ as

$$f^{(k)}(t) \sim \sum_{n \in \mathbb{Z}} d_n \mathrm{e}^{\mathrm{i}n\Omega t}, \qquad d_n = \frac{1}{T} \int_0^T f^{(k)}(t) \mathrm{e}^{-\mathrm{i}n\Omega t} \, \mathrm{d}t. \tag{1.52}$$

We can integrate this by parts to obtain

$$\begin{aligned} d_n &= \frac{1}{T} \left[f^{(k-1)}(t) \mathrm{e}^{-\mathrm{i}n\Omega t} \right]_0^T + \frac{1}{T} \int_0^T f^{(k-1)}(t) (\mathrm{i}\Omega n) \mathrm{e}^{-\mathrm{i}n\Omega t} \, \mathrm{d}t, \\ &= \frac{1}{T} (f^{(k-1)}(T) - f^{(k-1)}(0)) + \frac{\mathrm{i}\Omega n}{T} \int_0^T f^{(k-1)}(t) \mathrm{e}^{-\mathrm{i}n\Omega t} \, \mathrm{d}t, \\ &= \frac{\mathrm{i}\Omega n}{T} \int_0^T f^{(k-1)}(t) \mathrm{e}^{-\mathrm{i}n\Omega t} \, \mathrm{d}t \end{aligned} \tag{1.53}$$

where the boundary term vanished as $f^{(k-1)}$ is T-periodic. We can repeat this process until we reach

$$d_n = \frac{(\mathrm{i}\Omega n)^k}{T} \int_0^T f(t) \mathrm{e}^{-\mathrm{i}n\Omega t} \, \mathrm{d}t = (\mathrm{i}\Omega n)^k c_n. \tag{1.54}$$

Using part (i) of the previous proposition, we can bound $|d_n|$ by $\frac{1}{T} \int_0^T |f^{(k)}(t)| \, \mathrm{d}t$, which in turn allows us to bound $|c_n|$ as

$$|c_n| = \frac{|d_n|}{|\mathrm{i}\Omega n|^k} \leq \frac{1}{T(\Omega |n|)^k} \int_0^T |f^{(k)}(t)| \, \mathrm{d}t \doteq \frac{M}{|n|^k}, \tag{1.55}$$

giving the result. □

Now consider the case where f is a T-periodic function that is twice continuously differentiable. Then by proposition 1.2, we have

$$\left|\sum_{n\in\mathbb{Z}} c_n e^{i\Omega nt}\right| \leq |c_0| + \sum_{n=1}^{\infty}(|c_n| + |c_{-n}|) \leq |c_0| + 2M\sum_{n=1}^{\infty}\frac{1}{n^2}, \tag{1.56}$$

where $M = \frac{1}{T\Omega^2}\int_0^T |f''(t)|\,\mathrm{d}t$. As remarked earlier, the series $\sum_{n=1}^{\infty}\frac{1}{n^2}$ is convergent, so we conclude that the right-hand side is finite and the Fourier series for $f(t)$ is absolutely convergent.

1.6 Convergence of Fourier series

The previous section shows that under certain assumptions on the smoothness of f, the partial sums of a Fourier series converge absolutely to a finite number with a particular decay rate. However, if we relax the assumption of continuity, we don't necessarily have a divergent series; it could well be conditionally convergent. With this in mind, we now compare how the limit of the partial sums relates to the value of f at that point, with particular care taken at points of discontinuity.

Recall that the left and right limits of a function are defined to be

$$f(c^-) \doteq \lim_{t\to c^-} f(t), \qquad f(c^+) \doteq \lim_{t\to c^+} f(t), \tag{1.57}$$

respectively. That is, the left limit is given by approaching $t = c$ from the left, keeping $t < c$ as we approach it, while the right limit keeps $t > c$. For a continuous function, these two must match

$$f(c) = f(c^+) = f(c^-), \tag{1.58}$$

but for discontinuous functions they don't. In particular, if we plot a discontinuous function, we will see a jump of size $|f(c^+) - f(c^-)|$ at $t = c$. We can similarly define the left and right derivatives at a point, regardless of whether f is continuous. The *left derivative* at $t = c$ is given by

$$f'_-(c) = \lim_{h\to 0^-}\frac{f(c+h) - f(c^-)}{h}, \tag{1.59}$$

while the *right derivative* is defined as

$$f'_+(c) = \lim_{h\to 0^+}\frac{f(c+h) - f(c^+)}{h}. \tag{1.60}$$

Again, if f is continuous and differentiable, these two limits agree and we can unambiguously define $f'(c)$.

Example 1.8. For any $a \in \mathbb{R}$, consider the function

$$f(t) = \begin{cases} t^2, & t < 0, \\ t - a, & t \geq 0. \end{cases} \tag{1.61}$$

This is discontinuous at $t = 0$ if $a \neq 0$. Indeed, approaching from the left the limit is

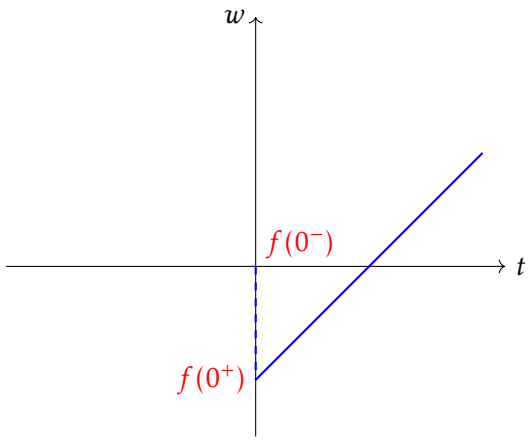


Figure 4: The function given in eq. (1.61) drawn with $a > 0$.

$f(0^-) = \lim_{t\to 0^-} t^2 = 0$, while the right limit is $f(0^+) = \lim_{t\to 0^+}(t-a) = -a$. The left and right derivatives are given by

$$f'_-(0) = \lim_{h\to 0^-} \frac{h^2 - 0}{h} = \lim_{h\to 0^-} h = 0, \tag{1.62}$$

and

$$f'_+(0) = \lim_{h\to 0^+} \frac{(h-a)+a}{h} = \lim_{h\to 0^+} \frac{h}{h} = 1. \tag{1.63}$$

Therefore, $f(t)$ is not differentiable at $t = 0$ for any value of a.

Equipped with this, we can now state *Dirichlet's criterion* for pointwise convergence of Fourier series.

Theorem 1.1. Let f be a T-periodic function and assume that $f'_+(t)$ and $f'_-(t)$ exist everywhere. Then the Fourier partial sums $S_N(t)$ converge pointwise to

$$\lim_{N\to\infty} S_N(t) = \frac{f(t^+) + f(t^-)}{2} = \lim_{\varepsilon\to 0^+} \frac{f(t+\varepsilon) + f(t-\varepsilon)}{2}, \tag{1.64}$$

for every $t \in \mathbb{R}$.

Remark. Note that Dirichlet's criterion has many different forms, so you might see various different formulations in other texts. A popular form in engineering textbooks is to require that f be integrable and have finitely many discontinuities and local extrema over a single period.

Given a function that satisfies the Dirichlet criterion, we can now replace the symbol $\sim$ with a genuine equality at points of continuity. On the other hand, at points of discontinuity the partial sums tend to the average of the left and right limits. We've actually already seen this in

action with the sawtooth wave. There, the Fourier series takes the form

$$f(t) \sim \sum_{n=1}^{\infty} \frac{2(-1)^{n+1}}{n} \sin nt, \tag{1.65}$$

with $t = \pi$ being a point of discontinuity. Substituting this into the Fourier series we get exactly zero since $\sin n\pi = 0$ for $n \in \mathbb{Z}$. Comparing this with the left and right limits, we see

$$f(\pi^-) = \pi, \quad f(\pi^+) = -\pi \implies \frac{f(\pi^+) + f(\pi^-)}{2} = 0, \tag{1.66}$$

matching the expectation from the Dirichlet criterion. Indeed, it is easy to see that the left and right derivatives are well defined at $t = \pi$, justifying the equality.

Example 1.9. Let's return to $f(t) = |\sin t|$ and investigate Fourier series convergence at $t = 0$. First note that $f(t)$ is continuous at $t = 0$, but not differentiable. Nonetheless, the left and right derivatives are well-defined and given by

$$f'_-(0) = \lim_{h \to 0^-} \frac{-\sin h}{h} = -1, \quad f'_+(0) = \lim_{h \to 0^+} \frac{\sin h}{h} = 1. \tag{1.67}$$

The final equalities can be justified by using the Maclaurin series for $\sin h$ or by using l'Hôpital's rule. Therefore, the Fourier series for $f(t)$ converges everywhere to $f(t)$ and we can write

$$f(t) = \frac{2}{\pi} - \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{\cos(2nt)}{4n^2 - 1}. \tag{1.68}$$

Taking $t = 0$ then gives us

$$0 = \frac{2}{\pi} - \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{1}{4n^2 - 1} \implies \sum_{n=1}^{\infty} \frac{1}{4n^2 - 1} = \frac{1}{2}. \tag{1.69}$$

1.7 L^2 theory and Parseval's theorem

So far, we've neglected an interesting property of periodic functions. Namely, if f and g are both T-periodic functions, then $\alpha f + \beta g$ is also T-periodic for any choice of constants α and β. This gives the space of T-periodic functions the structure of a *vector space* where the functions take on the role of 'vectors'. This begs the question, what other notions from linear algebra can we use to study T-periodic functions?

First, recall that for usual finite length vectors in $\mathbb{C}^n$, we have the standard inner product

$$\mathbf{u} \cdot \mathbf{v} = \sum_{i=1}^{n} u_i \overline{v}_i. \tag{1.70}$$

This inner product encodes important geometric data about a vector. For example, the squared length of a vector is given by

$$|\mathbf{u}|^2 = \mathbf{u} \cdot \mathbf{u} = \sum_{i=1}^{n} |u_i|^2. \tag{1.71}$$

Furthermore, it satisfies the *Cauchy-Schwarz inequality*

$$|\mathbf{u}\cdot\mathbf{v}| \leq |\mathbf{u}||\mathbf{v}|. \tag{1.72}$$

When two vectors satisfy $\mathbf{u}\cdot\mathbf{v}=0$, we say that $\mathbf{u}$ and $\mathbf{v}$ are *orthogonal*, extending the idea from real vector spaces that the two vectors lie on perpendicular lines through the origin. This is useful, as we can define an *orthonormal (ON) basis* to be a set of vectors $\{\mathbf{u}_i : i=1,\dots,n\}$ that satisfy

$$\mathbf{u}_i\cdot\mathbf{u}_j = \begin{cases} 1, & \text{if } i=j, \\ 0, & \text{if } i\neq j. \end{cases} \tag{1.73}$$

In doing so, we can expand any vector in $\mathbb{C}^n$ in terms of this basis

$$\mathbf{v} = \sum_{i=1}^{n} \alpha_i \mathbf{u}_i, \tag{1.74}$$

from which we want to calculate the constants α_i. Since the $\{\mathbf{u}_i\}$ form an orthonormal basis, we can take the inner product of $\mathbf{v}$ with $\mathbf{u}_m$ for some m to get

$$\mathbf{u}_m\cdot\mathbf{v} = \sum_{i=1}^{n} \alpha_i(\mathbf{u}_m\cdot\mathbf{u}_i) = \alpha_m, \tag{1.75}$$

where we've used the fact that $\mathbf{u}_m\cdot\mathbf{u}_i$ is only non-zero when $m=i$. So any vector can be easily expanded into an ON basis by calculating its inner products with the basis vectors.

This all seems fairly similar to Fourier series: we have some functions that we want to write as sums of simpler functions like $\mathrm{e}^{\mathrm{i}n\Omega t}$ and then we calculate the coefficients of this expansion by integrating $f(t)$ against these simple functions. We can make this more concrete by defining the inner product of two T-periodic functions to be

$$\langle f, g\rangle = \frac{1}{T}\int_0^T f(t)\overline{g(t)}\,\mathrm{d}t. \tag{1.76}$$

This is a natural extension of the inner product for vectors, now we just think about our 'vectors' as being indexed by the real numbers in $[0,T)$ and the summation is replaced by integration. If we now restrict to functions satisfying

$$\|f\|^2 \doteq \langle f, f\rangle = \frac{1}{T}\int_0^T |f(t)|^2\,\mathrm{d}t < \infty, \tag{1.77}$$

we obtain the so-called *Lebesgue space* $L^2([0,T])$. Equipping this space with the above inner product makes it into an *inner product space*. It is easy to check that

$$\langle \mathrm{e}^{\mathrm{i}m\Omega t}, \mathrm{e}^{\mathrm{i}n\Omega t}\rangle = \begin{cases} 1, & \text{if } m=n, \\ 0, & \text{if } m\neq n. \end{cases} \tag{1.78}$$

This signals that the functions $f_m(t) = \mathrm{e}^{\mathrm{i}m\Omega t}$ act like an ON basis for $L^2([0,T])$, though mathematically we should be a little more careful. We haven't established that it is a complete basis and haven't dealt with the nuances of having an infinite set of basis vectors, but we will be cavalier and assume things are fine. The bottom line is that expanding a T-periodic function that belongs to $L^2([0,T])$ into complex exponentials is essentially just an ON basis expansion where the Fourier coefficients are given by

$$c_m \doteq \langle f(t), \mathrm{e}^{\mathrm{i}m\Omega t}\rangle = \frac{1}{T}\int_0^T f(t)\mathrm{e}^{-\mathrm{i}m\Omega t}\,\mathrm{d}t. \tag{1.79}$$

Similar statements hold for the trigonometric expansions, it is just a basis change from the exponential basis.

The quantity $\|f\|^2$ is the analogue of the length of a vector. In that case, instead of measuring the length directly, we expand into an ON basis and then use Pythagoras's theorem. We can do the same in $L^2([0,T])$, it goes by the name of *Parseval's theorem.*

Theorem 1.2. If $f \in L^2([0,T])$ has Fourier series

$$f(t) \sim \sum_{n\in\mathbb{Z}} c_n \mathrm{e}^{\mathrm{i}n\Omega t} = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n\cos(n\Omega t) + b_n\sin(n\Omega t), \tag{1.80}$$

then

$$\frac{1}{T}\int_0^T |f(t)|^2\,\mathrm{d}t = \sum_{n\in\mathbb{Z}}|c_n|^2 = \frac{|a_0|^2}{4} + \frac{1}{2}\sum_{n=1}^{\infty}\Big(|a_n|^2 + |b_n|^2\Big). \tag{1.81}$$

Proof. Substituting the Fourier expansion of $\overline{f(t)}$ into $\|f\|^2$ gives

$$\begin{aligned}\|f\|^2 &= \frac{1}{T}\int_0^T f(t)\left(\sum_{n\in\mathbb{Z}}\overline{c}_n\mathrm{e}^{-\mathrm{i}n\Omega t}\right)\mathrm{d}t,\\ &= \sum_{n\in\mathbb{Z}}\frac{\overline{c}_n}{T}\int_0^T f(t)\mathrm{e}^{-\mathrm{i}n\Omega t}\,\mathrm{d}t,\end{aligned} \tag{1.82}$$

where we've assumed that we can switch integration and summation. We now recognise the integral as the definition of $c_m = \langle f(t), \mathrm{e}^{\mathrm{i}m\Omega t}\rangle$, so the sum becomes

$$\|f\|^2 = \sum_{n\in\mathbb{Z}}\overline{c}_n c_n = \sum_{n\in\mathbb{Z}}|c_n|^2. \tag{1.83}$$

Rewriting the c_n in terms of a_n and b_n and expanding the squares then gives the trigonometric result. □

Example 1.10. The sawtooth function belongs to $L^2([0,T])$, so we can use Parseval's theorem.

On one side we have

$$\|f\|^2 = \frac{1}{2\pi}\int_{-\pi}^{\pi} t^2\,\mathrm{d}t = \frac{1}{2\pi}\left[\frac{t^3}{3}\right]_{-\pi}^{\pi} = \frac{\pi^2}{3}. \tag{1.84}$$

The Fourier series is given by eq. (1.27), giving the sum

$$\sum_{n\in\mathbb{Z}} |c_n|^2 = \sum_{n\neq 0}\frac{1}{n^2} = 2\sum_{n=1}^{\infty}\frac{1}{n^2}. \tag{1.85}$$

Equating both results and dividing by two then gives

$$\sum_{n=1}^{\infty}\frac{1}{n^2} = \frac{\pi^2}{6}. \tag{1.86}$$

This is the famous solution to the Basel problem, first solved by Euler in 1734.

1.8 The Gibbs phenomenon ★

— Work in progress, very non-examinable —